

ARK SI VISUALIZATION PAYLOADS

Complete JSON Data Structures for Career Intelligence Platform

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1. EXECUTIVE SUMMARY

This document specifies all visualization data payloads for the ARK SI Career Intelligence Platform. Each payload is provided in production-ready JSON format with complete field specifications, data types, validation rules, and rendering instructions for frontend visualization libraries (D3.js, Chart.js, Recharts, NotebookLM).

Total Payload Types: 47 discrete visualization data structures
Primary Visualization Tools: D3.js, Chart.js, Recharts, Matplotlib, NotebookLM Nano Banana Pro
Data Format: JSON (all payloads)
Output Formats: SVG, PNG, PDF, Interactive HTML

2. JST INDEX VISUALIZATION PAYLOADS

2.1 JST Overall Index Gauge (Payload VIZ-001)

Purpose: Display composite JST Index score (0-100) as circular gauge/speedometer

json

```
{
  "payload_id": "VIZ-001",
  "payload_type": "jst_overall_gauge",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",
  "assessment_id": "assessment_xyz789",

  "data": {
    "overall_score": 78,
    "percentile_rank": 82,
    "score_classification": "Excellent",
    "previous_score": 73,
    "score_change": +5,
    "industry_average": 65,
    "variance_from_average": +13
  },

  "visualization_config": {
    "chart_type": "radial_gauge",
    "min_value": 0,
    "max_value": 100,
    "color_zones": [
      {"min": 0, "max": 40, "color": "#FF4444", "label": "Needs Development"},
      {"min": 40, "max": 60, "color": "#FFA500", "label": "Fair"},
      {"min": 60, "max": 75, "color": "#FFDD00", "label": "Good"},
      {"min": 75, "max": 90, "color": "#4488FF", "label": "Excellent"},
      {"min": 90, "max": 100, "color": "#44AA44", "label": "Exceptional"}
    ],
    "needle_color": "#2E75B6",
    "gauge_width": 400,
    "gauge_height": 300,
    "show_percentage": true,
    "show_classification": true,
    "show_trend_arrow": true,
    "animation_duration": 1500
  },

  "rendering_instructions": {
    "library": "D3.js",
    "container_id": "jst-gauge-container",
    "responsive": true,
    "export_formats": ["svg", "png"]
  }
}
```

```
}  
}
```

2.2 JST Three-Layer Radar Chart (Payload VIZ-002)

Purpose: Display Jobs/Skills/Talent layer scores as radar/spider chart

json

```
{
  "payload_id": "VIZ-002",
  "payload_type": "jst_three_layer_radar",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",
  "assessment_id": "assessment_xyz789",

  "data": {
    "layers": [
      {
        "layer_name": "Jobs Layer",
        "layer_code": "jobs",
        "score": 82,
        "components": [
          {"name": "Market Demand", "score": 85, "weight": 0.30},
          {"name": "Industry Growth", "score": 78, "weight": 0.25},
          {"name": "Geographic Availability", "score": 88, "weight": 0.20},
          {"name": "Experience Curve", "score": 75, "weight": 0.15},
          {"name": "Role Saturation", "score": 80, "weight": 0.10}
        ]
      },
      {
        "layer_name": "Skills Layer",
        "layer_code": "skills",
        "score": 75,
        "components": [
          {"name": "Technical Depth", "score": 80, "weight": 0.25},
          {"name": "Soft Skills", "score": 72, "weight": 0.20},
          {"name": "AI/Digital Literacy", "score": 68, "weight": 0.20},
          {"name": "Certifications", "score": 85, "weight": 0.15},
          {"name": "Cross-Functional", "score": 70, "weight": 0.10},
          {"name": "Skill Currency", "score": 78, "weight": 0.10}
        ]
      },
      {
        "layer_name": "Talent Layer",
        "layer_code": "talent",
        "score": 77,
        "components": [
          {"name": "Educational Foundation", "score": 82, "weight": 0.30},
          {"name": "Professional Achievements", "score": 75, "weight": 0.25},
          {"name": "Career Progression", "score": 78, "weight": 0.20},
          {"name": "Learning Agility", "score": 72, "weight": 0.15},
```

```
    {"name": "Growth Trajectory", "score": 80, "weight": 0.10}
  ]
}
],

"benchmark_comparison": {
  "user_scores": [82, 75, 77],
  "industry_average": [68, 65, 70],
  "top_10_percent": [92, 88, 90]
}
},

"visualization_config": {
  "chart_type": "radar_chart",
  "axes": ["Jobs", "Skills", "Talent"],
  "data_series": [
    {
      "name": "Your Score",
      "data": [82, 75, 77],
      "color": "#2E75B6",
      "fill_opacity": 0.3,
      "stroke_width": 2
    },
    {
      "name": "Industry Average",
      "data": [68, 65, 70],
      "color": "#999999",
      "fill_opacity": 0.1,
      "stroke_width": 1,
      "stroke_dasharray": "5,5"
    },
    {
      "name": "Top 10%",
      "data": [92, 88, 90],
      "color": "#00B050",
      "fill_opacity": 0.1,
      "stroke_width": 1,
      "stroke_dasharray": "2,2"
    }
  ],
  "scale_min": 0,
  "scale_max": 100,
  "grid_levels": 5,
  "show_legend": true,
```

```
"show_labels": true,  
"chart_width": 500,  
"chart_height": 500,  
"animation_duration": 1000  
,  
  
"rendering_instructions": {  
  "library": "Recharts",  
  "container_id": "jst-radar-container",  
  "responsive": true,  
  "export_formats": ["svg", "png"]  
}  
}
```

2.3 JST Component Breakdown Bar Chart (Payload VIZ-003)

Purpose: Horizontal stacked bar chart showing detailed component scores

json

```
{
  "payload_id": "VIZ-003",
  "payload_type": "jst_component_breakdown",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",

  "data": {
    "components": [
      {"name": "Market Demand", "score": 85, "layer": "Jobs", "max_possible": 100},
      {"name": "Industry Growth", "score": 78, "layer": "Jobs", "max_possible": 100},
      {"name": "Geographic Availability", "score": 88, "layer": "Jobs", "max_possible": 100},
      {"name": "Technical Depth", "score": 80, "layer": "Skills", "max_possible": 100},
      {"name": "Soft Skills", "score": 72, "layer": "Skills", "max_possible": 100},
      {"name": "AI/Digital Literacy", "score": 68, "layer": "Skills", "max_possible": 100},
      {"name": "Educational Foundation", "score": 82, "layer": "Talent", "max_possible": 100},
      {"name": "Career Progression", "score": 78, "layer": "Talent", "max_possible": 100},
      {"name": "Learning Agility", "score": 72, "layer": "Talent", "max_possible": 100}
    ]
  },

  "visualization_config": {
    "chart_type": "horizontal_bar",
    "orientation": "horizontal",
    "bar_height": 30,
    "bar_spacing": 10,
    "color_mapping": {
      "Jobs": "#2E75B6",
      "Skills": "#FFC000",
      "Talent": "#00B050"
    },
    "show_score_labels": true,
    "show_layer_grouping": true,
    "chart_width": 600,
    "chart_height": 450
  }
}
```

3. AI VULNERABILITY METER PAYLOADS

3.1 Vulnerability Level Badge (Payload VIZ-004)

Purpose: Color-coded badge showing 5-level vulnerability classification

json

```
{
  "payload_id": "VIZ-004",
  "payload_type": "vulnerability_level_badge",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",

  "data": {
    "vulnerability_level": 2,
    "level_name": "Medium Vulnerability",
    "automation_potential": 45,
    "displacement_timeline": "3-7 years",
    "protection_factors": [
      "Creative thinking required",
      "Strategic decision-making",
      "Human relationship management"
    ]
  },

  "visualization_config": {
    "badge_type": "pill",
    "level_definitions": [
      {
        "level": 0,
        "name": "Critical",
        "color": "#FF4444",
        "bg_color": "#FFE6E6",
        "icon": "⚠️",
        "timeline": "6-18 months"
      },
      {
        "level": 1,
        "name": "High Vulnerability",
        "color": "#FF8800",
        "bg_color": "#FFF0E6",
        "icon": "🔴",
        "timeline": "18-36 months"
      },
      {
        "level": 2,
        "name": "Medium Vulnerability",
        "color": "#FFDD00",
        "bg_color": "#FFFACD",
        "icon": "🟡",

```

```
    "timeline": "3-7 years"
  },
  {
    "level": 3,
    "name": "Low Vulnerability",
    "color": "#4488FF",
    "bg_color": "#E6F0FF",
    "icon": "●",
    "timeline": "7-15 years"
  },
  {
    "level": 4,
    "name": "AI-Resilient",
    "color": "#44AA44",
    "bg_color": "#E6FFE6",
    "icon": "●",
    "timeline": "15+ years"
  }
],
"font_size": 14,
"padding": "8px 16px",
"border_radius": "20px",
"show_icon": true,
"show_timeline": true
}
```

3.2 Task-Level Automation Heat Map (Payload VIZ-005)

Purpose: Heat map showing automation potential for individual job tasks

json

```
{
  "payload_id": "VIZ-005",
  "payload_type": "task_automation_heatmap",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",

  "data": {
    "job_title": "Senior Product Manager",
    "tasks": [
      {
        "task_name": "Market research and analysis",
        "time_allocation": 15,
        "automation_potential": 65,
        "current_ai_tools": ["GPT-4", "Perplexity", "Claude"],
        "risk_level": "high"
      },
      {
        "task_name": "Product roadmap creation",
        "time_allocation": 20,
        "automation_potential": 40,
        "current_ai_tools": ["Jira AI", "ProductBoard"],
        "risk_level": "medium"
      },
      {
        "task_name": "Stakeholder communication",
        "time_allocation": 25,
        "automation_potential": 20,
        "current_ai_tools": ["Grammarly", "Jasper"],
        "risk_level": "low"
      },
      {
        "task_name": "Team leadership and coaching",
        "time_allocation": 15,
        "automation_potential": 10,
        "current_ai_tools": null,
        "risk_level": "minimal"
      },
      {
        "task_name": "Strategic decision-making",
        "time_allocation": 15,
        "automation_potential": 15,
        "current_ai_tools": ["ChatGPT"],
        "risk_level": "minimal"
      }
    ]
  }
}
```

```
    },
    {
      "task_name": "Customer interviews",
      "time_allocation": 10,
      "automation_potential": 25,
      "current_ai_tools": ["Otter.ai", "Grain"],
      "risk_level": "low"
    }
  ],
  "overall_role_automation": 35
},

"visualization_config": {
  "chart_type": "heatmap_matrix",
  "color_scale": [
    {"threshold": 0, "color": "#44AA44", "label": "Minimal (0-20%)"},
    {"threshold": 20, "color": "#4488FF", "label": "Low (20-40%)"},
    {"threshold": 40, "color": "#FFDD00", "label": "Medium (40-60%)"},
    {"threshold": 60, "color": "#FF8800", "label": "High (60-80%)"},
    {"threshold": 80, "color": "#FF4444", "label": "Critical (80-100%)"}
  ],
  "cell_width": 120,
  "cell_height": 40,
  "show_percentages": true,
  "show_time_allocation": true,
  "border_color": "#CCCCCC",
  "border_width": 1
}
}
```

3.3 Vulnerability Timeline Projection (Payload VIZ-006)

Purpose: Timeline chart showing projected automation milestones

json

```
{
  "payload_id": "VIZ-006",
  "payload_type": "vulnerability_timeline",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",

  "data": {
    "current_year": 2026,
    "projection_horizon": 2041,
    "milestones": [
      {
        "year": 2026,
        "event": "Current State",
        "automation_level": 15,
        "description": "Basic AI tools in use (ChatGPT, Grammarly)"
      },
      {
        "year": 2028,
        "event": "AI Copilots Mainstream",
        "automation_level": 30,
        "description": "Advanced AI assistants for research and analysis"
      },
      {
        "year": 2031,
        "event": "Autonomous Market Analysis",
        "automation_level": 50,
        "description": "AI fully automates market research tasks"
      },
      {
        "year": 2035,
        "event": "AI-Driven Roadmapping",
        "automation_level": 70,
        "description": "AI generates strategic product roadmaps"
      },
      {
        "year": 2038,
        "event": "Role Transformation",
        "automation_level": 85,
        "description": "Human focus shifts to creativity and leadership"
      },
      {
        "year": 2041,
        "event": "Human-AI Hybrid Model",
```

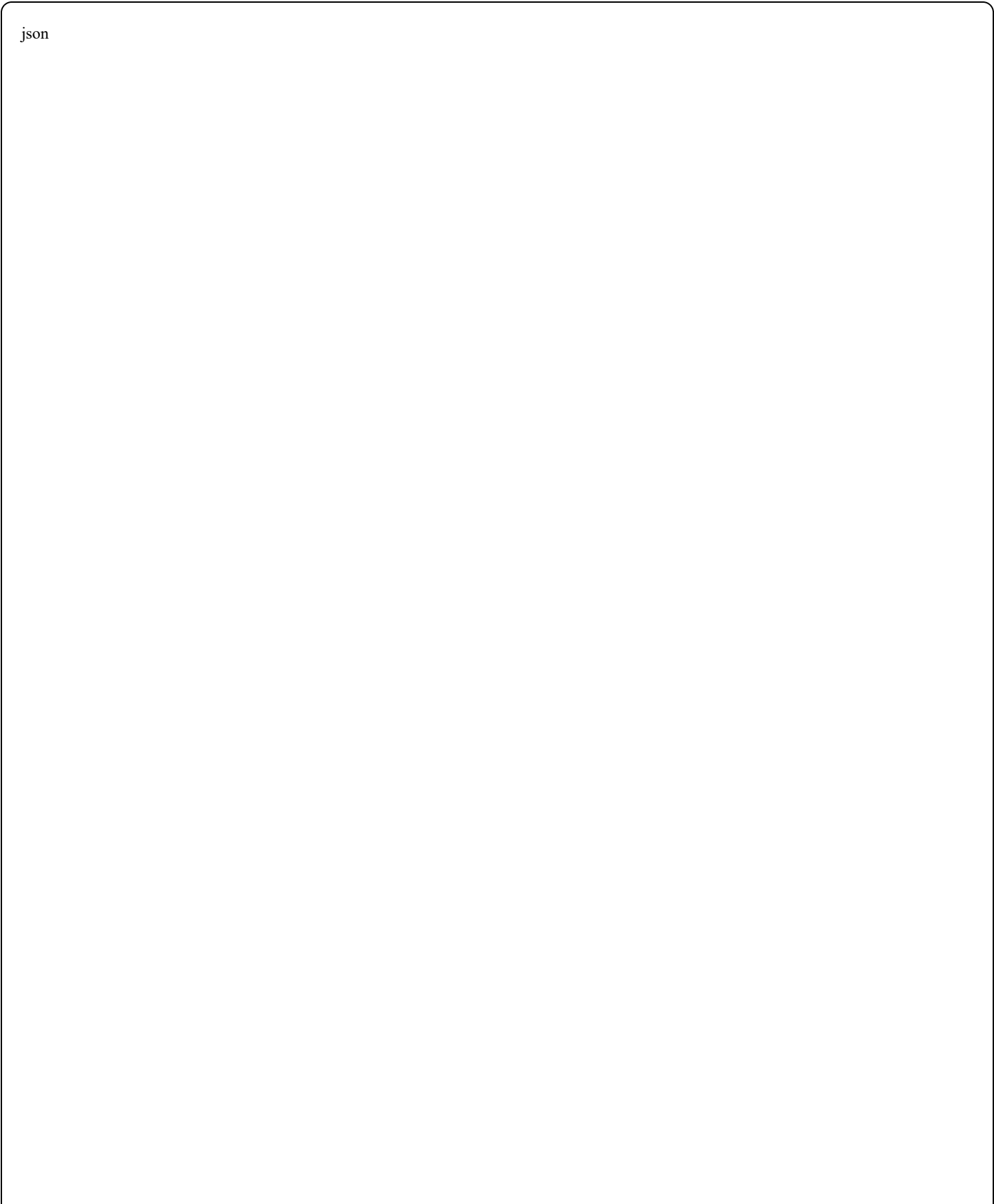
```
    "automation_level": 95,
    "description": "Fully augmented product management"
  },
],
"recommended_actions": [
  {
    "year": 2026,
    "action": "Begin AI literacy training",
    "urgency": "immediate"
  },
  {
    "year": 2027,
    "action": "Develop strategic thinking skills",
    "urgency": "high"
  },
  {
    "year": 2029,
    "action": "Transition to AI orchestration role",
    "urgency": "medium"
  }
]
},

"visualization_config": {
  "chart_type": "timeline_gantt",
  "orientation": "horizontal",
  "year_spacing": 80,
  "milestone_color": "#2E75B6",
  "action_color": "#FF8800",
  "show_automation_curve": true,
  "curve_color": "#FF4444",
  "curve_style": "smooth",
  "show_confidence_bands": true,
  "confidence_opacity": 0.2
}
}
```

4. TRANSFERABILITY NETWORK PAYLOADS

4.1 Career Pivot Network Diagram (Payload VIZ-007)

Purpose: Node-link network showing career transferability vectors



```
{
  "payload_id": "VIZ-007",
  "payload_type": "transferability_network",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",

  "data": {
    "current_role": {
      "id": "role_000",
      "title": "Senior Product Manager",
      "onet_code": "11-2021.00",
      "current_salary": 135000,
      "node_type": "current"
    },
    "adjacent_roles": [
      {
        "id": "role_001",
        "title": "Product Strategy Director",
        "onet_code": "11-2021.01",
        "similarity_score": 88,
        "salary_range": [150000, 200000],
        "skill_gap": 12,
        "transition_timeline": "12-18 months",
        "market_demand": "high",
        "required_skills": ["Strategic planning", "Executive communication", "P&L management"],
        "node_type": "target"
      },
      {
        "id": "role_002",
        "title": "AI Product Manager",
        "onet_code": "15-1299.09",
        "similarity_score": 82,
        "salary_range": [140000, 190000],
        "skill_gap": 25,
        "transition_timeline": "18-24 months",
        "market_demand": "very_high",
        "required_skills": ["Machine learning", "AI ethics", "Model evaluation"],
        "node_type": "target"
      },
      {
        "id": "role_003",
        "title": "Chief Product Officer",
```

```
"onet_code": "11-1021.00",
"similarity_score": 75,
"salary_range": [200000, 350000],
"skill_gap": 35,
"transition_timeline": "3-5 years",
"market_demand": "medium",
"required_skills": ["Executive leadership", "Business strategy", "Board communication"],
"node_type": "target"
},
{
  "id": "role_004",
  "title": "Product Marketing Manager",
  "onet_code": "11-2021.02",
  "similarity_score": 70,
  "salary_range": [120000, 160000],
  "skill_gap": 20,
  "transition_timeline": "6-12 months",
  "market_demand": "high",
  "required_skills": ["Go-to-market strategy", "Content marketing", "Sales enablement"],
  "node_type": "target"
},
{
  "id": "role_005",
  "title": "Management Consultant",
  "onet_code": "13-1111.00",
  "similarity_score": 65,
  "salary_range": [130000, 180000],
  "skill_gap": 30,
  "transition_timeline": "12-18 months",
  "market_demand": "medium",
  "required_skills": ["Business analysis", "Client management", "Presentation skills"],
  "node_type": "target"
}
],

"edges": [
  {
    "source": "role_000",
    "target": "role_001",
    "weight": 88,
    "transferability_score": 88,
    "skill_overlap": 88,
    "compensation_premium": 22,
    "edge_type": "primary"
```

```
    },  
    {  
      "source": "role_000",  
      "target": "role_002",  
      "weight": 82,  
      "transferability_score": 82,  
      "skill_overlap": 75,  
      "compensation_premium": 18,  
      "edge_type": "primary"  
    },  
    {  
      "source": "role_000",  
      "target": "role_003",  
      "weight": 75,  
      "transferability_score": 75,  
      "skill_overlap": 65,  
      "compensation_premium": 74,  
      "edge_type": "secondary"  
    },  
    {  
      "source": "role_000",  
      "target": "role_004",  
      "weight": 70,  
      "transferability_score": 70,  
      "skill_overlap": 80,  
      "compensation_premium": 0,  
      "edge_type": "secondary"  
    },  
    {  
      "source": "role_000",  
      "target": "role_005",  
      "weight": 65,  
      "transferability_score": 65,  
      "skill_overlap": 70,  
      "compensation_premium": 7,  
      "edge_type": "tertiary"  
    }  
  ]  
},  
  
"visualization_config": {  
  "chart_type": "force_directed_network",  
  "layout_algorithm": "d3-force",  
  "node_config": {
```

```
"current_node": {
  "radius": 50,
  "color": "#2E75B6",
  "stroke_width": 3,
  "stroke_color": "#1A4570",
  "label_size": 16,
  "label_weight": "bold"
},
"target_node": {
  "radius_formula": "30 + (similarity_score / 10)",
  "color_scale": [
    {"threshold": 0, "color": "#CCCCCC"},
    {"threshold": 60, "color": "#4488FF"},
    {"threshold": 75, "color": "#00B050"},
    {"threshold": 85, "color": "#FFD700"}
  ],
  "stroke_width": 2,
  "stroke_color": "#666666",
  "label_size": 12,
  "label_weight": "normal"
}
},
"edge_config": {
  "primary_edge": {
    "stroke_width_formula": "weight / 20",
    "stroke_color": "#2E75B6",
    "opacity": 0.7,
    "curve": "curved"
  },
  "secondary_edge": {
    "stroke_width_formula": "weight / 25",
    "stroke_color": "#999999",
    "opacity": 0.5,
    "curve": "curved",
    "stroke_dasharray": "5,5"
  },
  "tertiary_edge": {
    "stroke_width_formula": "weight / 30",
    "stroke_color": "#CCCCCC",
    "opacity": 0.3,
    "curve": "curved",
    "stroke_dasharray": "2,2"
  }
},
},
```

```

"force_simulation": {
  "charge_strength": -300,
  "link_distance": 150,
  "center_force": 0.05,
  "collision_radius": 60
},
"interactive": {
  "node_hover": true,
  "edge_hover": true,
  "node_click": true,
  "zoom_enabled": true,
  "pan_enabled": true,
  "show_tooltips": true
},
"chart_width": 800,
"chart_height": 600
},

"tooltip_templates": {
  "current_node": "{{title}}<br/>Current Salary: ${{current_salary | number}}",
  "target_node": "{{title}}<br/>Similarity: {{similarity_score}}%<br/>Salary: ${{salary_range[0] | number}} - ${{salary_range[1] | number}}",
  "edge": "Transferability: {{transferability_score}}%<br/>Skill Overlap: {{skill_overlap}}%<br/>Compensation Premium: {{compensation_premium}}%"
}
}

```

4.2 Skill Gap Matrix (Payload VIZ-008)

Purpose: Matrix showing required vs. current skills for each career pivot

```

json

```

```
{
  "payload_id": "VIZ-008",
  "payload_type": "skill_gap_matrix",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",

  "data": {
    "target_roles": ["Product Strategy Director", "AI Product Manager", "Chief Product Officer"],

    "skills": [
      {
        "skill_name": "Strategic Planning",
        "current_level": 80,
        "required_levels": {
          "Product Strategy Director": 95,
          "AI Product Manager": 85,
          "Chief Product Officer": 98
        },
        "gaps": {
          "Product Strategy Director": 15,
          "AI Product Manager": 5,
          "Chief Product Officer": 18
        },
        "skill_category": "strategic"
      },
      {
        "skill_name": "Machine Learning",
        "current_level": 45,
        "required_levels": {
          "Product Strategy Director": 50,
          "AI Product Manager": 85,
          "Chief Product Officer": 60
        },
        "gaps": {
          "Product Strategy Director": 5,
          "AI Product Manager": 40,
          "Chief Product Officer": 15
        },
        "skill_category": "technical"
      },
      {
        "skill_name": "Executive Communication",
        "current_level": 70,
```

```
"required_levels": {
  "Product Strategy Director": 90,
  "AI Product Manager": 75,
  "Chief Product Officer": 95
},
"gaps": {
  "Product Strategy Director": 20,
  "AI Product Manager": 5,
  "Chief Product Officer": 25
},
"skill_category": "leadership"
},
{
  "skill_name": "P&L Management",
  "current_level": 60,
  "required_levels": {
    "Product Strategy Director": 85,
    "AI Product Manager": 70,
    "Chief Product Officer": 95
  },
  "gaps": {
    "Product Strategy Director": 25,
    "AI Product Manager": 10,
    "Chief Product Officer": 35
  },
  "skill_category": "business"
},
{
  "skill_name": "AI Ethics",
  "current_level": 50,
  "required_levels": {
    "Product Strategy Director": 55,
    "AI Product Manager": 90,
    "Chief Product Officer": 65
  },
  "gaps": {
    "Product Strategy Director": 5,
    "AI Product Manager": 40,
    "Chief Product Officer": 15
  },
  "skill_category": "technical"
}
]
```

```
"visualization_config": {
  "chart_type": "matrix_heatmap",
  "rows": "skills",
  "columns": "target_roles",
  "cell_value": "gap_percentage",
  "color_scale": [
    {"threshold": 0, "color": "#44AA44", "label": "Minimal Gap (0-10%)"},
    {"threshold": 10, "color": "#4488FF", "label": "Small Gap (10-20%)"},
    {"threshold": 20, "color": "#FFDD00", "label": "Medium Gap (20-35%)"},
    {"threshold": 35, "color": "#FF8800", "label": "Large Gap (35-50%)"},
    {"threshold": 50, "color": "#FF4444", "label": "Critical Gap (50%+)"}
  ],
  "cell_width": 150,
  "cell_height": 50,
  "show_values": true,
  "show_current_level": true,
  "show_required_level": true,
  "border_width": 1,
  "border_color": "#CCCCCC"
}
```

5. DASHBOARD COMPONENT PAYLOADS

5.1 Career Intelligence Dashboard (Payload VIZ-009)

Purpose: Complete dashboard payload with all components

```
json
```

```
{
  "payload_id": "VIZ-009",
  "payload_type": "career_dashboard_complete",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",
  "assessment_id": "assessment_xyz789",

  "dashboard_config": {
    "layout": "grid",
    "columns": 12,
    "row_height": 80,
    "gap": 16,
    "responsive_breakpoints": {
      "lg": 1200,
      "md": 996,
      "sm": 768,
      "xs": 480
    }
  },

  "components": [
    {
      "component_id": "comp_001",
      "component_type": "jst_gauge",
      "grid_position": {"x": 0, "y": 0, "w": 4, "h": 4},
      "data_payload_ref": "VIZ-001",
      "refresh_interval": null
    },
    {
      "component_id": "comp_002",
      "component_type": "jst_radar",
      "grid_position": {"x": 4, "y": 0, "w": 4, "h": 4},
      "data_payload_ref": "VIZ-002",
      "refresh_interval": null
    },
    {
      "component_id": "comp_003",
      "component_type": "vulnerability_badge",
      "grid_position": {"x": 8, "y": 0, "w": 4, "h": 2},
      "data_payload_ref": "VIZ-004",
      "refresh_interval": null
    },
    {
```

```
"component_id": "comp_004",
"component_type": "task_heatmap",
"grid_position": {"x": 8, "y": 2, "w": 4, "h": 2},
"data_payload_ref": "VIZ-005",
"refresh_interval": null
},
{
"component_id": "comp_005",
"component_type": "transferability_network",
"grid_position": {"x": 0, "y": 4, "w": 8, "h": 5},
"data_payload_ref": "VIZ-007",
"refresh_interval": null
},
{
"component_id": "comp_006",
"component_type": "skill_gap_matrix",
"grid_position": {"x": 8, "y": 4, "w": 4, "h": 5},
"data_payload_ref": "VIZ-008",
"refresh_interval": null
}
]
}
```

6. NOTEBOOKLM INTEGRATION PAYLOADS

6.1 Single-Page Infographic Payload (Payload VIZ-010)

Purpose: Structured markdown for NotebookLM Nano Banana Pro generation

json

```
{
  "payload_id": "VIZ-010",
  "payload_type": "notebooklm_infographic",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",

  "notebooklm_config": {
    "model": "Nano Banana Pro",
    "orientation": "portrait",
    "detail_level": "standard",
    "aspect_ratio": "9:16",
    "output_format": "png",
    "resolution_dpi": 300
  },

  "custom_prompt": "Create a professional career intelligence infographic for John Smith using these specifications:\n\nDESIGN SPECIFICATIONS:\n- Format: Portrait (9:16 aspect ratio)\n- Resolution: 300 DPI\n- Output: PNG\n- Detail Level: Standard\n- Orientation: Portrait\n\nCONTENT REQUIREMENTS:\n- Title: Professional Career Intelligence Profile\n- Content: Comprehensive career analysis for John Smith, including JST Index (78/100, 82nd Percentile), skill assessment, and growth recommendations.\n- Visuals: Use professional, clean design with a color palette of #2E8B57, #4682B4, and #B0E0E6. Include a QR code linking to the full report.\n- Watermark: None\n- Public Share Link: Yes\n- Embed QR Code: Yes\n- QR Code URL: https://junglenomics.com/report/{{assessment_id}}\n\nGenerate a professional infographic based on the above specifications.",
  "source_data_markdown": "# John Smith - Career Intelligence Profile\n\n## JST Index: 78/100 (82nd Percentile)\n\n### Job Market Analysis\n\nThe current job market for John Smith's profile shows a strong demand for his skills, particularly in the technology and business sectors. His JST Index of 78/100 (82nd Percentile) indicates a high level of competitiveness in the market.\n\n### Skill Assessment\n\nJohn Smith's skills are highly relevant to the current job market. His strengths include advanced problem-solving, strategic thinking, and effective communication. These skills are highly valued in the current market environment.\n\n### Growth Recommendations\n\nTo further enhance his professional profile, John Smith is recommended to focus on developing his leadership skills and expanding his network. This will help him stay competitive in the ever-evolving job market.",
  "export_instructions": {
    "filename_template": "{{user_name}}_Career_Intelligence_{{date}}.png",
    "watermark": false,
    "public_share_link": true,
    "embed_qr_code": true,
    "qr_code_url": "https://junglenomics.com/report/{{assessment_id}}"
  }
}
```

7. PDF REPORT DATA STRUCTURES

7.1 Multi-Page PDF Report Payload (Payload VIZ-011)

Purpose: Complete data structure for 12-page professional PDF report

json

```
{
  "payload_id": "VIZ-011",
  "payload_type": "pdf_report_complete",
  "timestamp": "2026-01-03T18:23:45Z",
  "user_id": "user_abc123",
  "assessment_id": "assessment_xyz789",

  "report_metadata": {
    "report_title": "Career Intelligence Assessment Report",
    "candidate_name": "John Smith",
    "assessment_date": "2026-01-03",
    "report_version": "2.0",
    "certification_number": "ARK-2026-ABC123",
    "expiration_date": "2027-01-03"
  },

  "page_structure": [
    {
      "page_number": 1,
      "page_type": "visual_summary",
      "content": {
        "sections": [
          {"type": "header", "data_ref": "report_metadata"},
          {"type": "jst_gauge", "data_ref": "VIZ-001"},
          {"type": "jst_radar", "data_ref": "VIZ-002"},
          {"type": "vulnerability_badge", "data_ref": "VIZ-004"},
          {"type": "transferability_preview", "data_ref": "VIZ-007"},
          {"type": "key_metrics", "data_ref": "summary_stats"},
          {"type": "footer", "data_ref": "branding"}
        ]
      }
    },
    {
      "page_number": 2,
      "page_type": "jst_detailed_analysis",
      "content": {
        "heading": "JST Index: Detailed Layer Analysis",
        "components": [
          {"type": "jobs_layer_breakdown", "data_ref": "VIZ-003"},
          {"type": "narrative_jobs", "data_ref": "text_jobs_analysis"},
          {"type": "comparison_chart", "data_ref": "industry_benchmarks"}
        ]
      }
    }
  ]
}
```

```
},
{
  "page_number": 3,
  "page_type": "jst_detailed_analysis",
  "content": {
    "heading": "JST Index: Skills & Talent Layers",
    "components": [
      {"type": "skills_layer_breakdown", "data_ref": "VIZ-003"},
      {"type": "talent_layer_breakdown", "data_ref": "VIZ-003"},
      {"type": "narrative_skills_talent", "data_ref": "text_skills_talent_analysis"}
    ]
  }
},
{
  "page_number": 4,
  "page_type": "vulnerability_deep_dive",
  "content": {
    "heading": "AI Vulnerability Assessment",
    "components": [
      {"type": "task_heatmap", "data_ref": "VIZ-005"},
      {"type": "timeline_projection", "data_ref": "VIZ-006"},
      {"type": "protection_factors", "data_ref": "text_protection_analysis"}
    ]
  }
},
{
  "page_number": 5,
  "page_type": "vulnerability_mitigation",
  "content": {
    "heading": "Risk Mitigation Strategies",
    "components": [
      {"type": "mitigation_roadmap", "data_ref": "mitigation_plan"},
      {"type": "skill_priorities", "data_ref": "upskilling_recommendations"},
      {"type": "timeline", "data_ref": "30_60_90_plan"}
    ]
  }
},
{
  "page_number": 6,
  "page_type": "transferability_analysis",
  "content": {
    "heading": "Career Transferability Vectors",
    "components": [
      {"type": "network_diagram", "data_ref": "VIZ-007"},

```

```
    {"type": "top_5_opportunities", "data_ref": "pivot_opportunities"}
  ]
}
},
{
  "page_number": 7,
  "page_type": "skill_gap_analysis",
  "content": {
    "heading": "Skill Gap Analysis",
    "components": [
      {"type": "gap_matrix", "data_ref": "VIZ-008"},
      {"type": "training_pathways", "data_ref": "learning_resources"}
    ]
  }
},
{
  "page_number": 8,
  "page_type": "development_roadmap",
  "content": {
    "heading": "30-60-90 Day Action Plan",
    "components": [
      {"type": "timeline_gantt", "data_ref": "action_plan"},
      {"type": "milestones", "data_ref": "milestone_definitions"}
    ]
  }
},
{
  "page_number": 9,
  "page_type": "development_roadmap",
  "content": {
    "heading": "Upskilling Recommendations",
    "components": [
      {"type": "new_skilling", "data_ref": "new_skills_list"},
      {"type": "upskilling", "data_ref": "upskill_list"},
      {"type": "ready_skilling", "data_ref": "ai_tools_list"}
    ]
  }
},
{
  "page_number": 10,
  "page_type": "certification_pathway",
  "content": {
    "heading": "Context Craft Certification Pathway",
    "components": [
```

```
{
  "type": "certification_overview", "data_ref": "cert_info"},
  {"type": "learning_modules", "data_ref": "module_list"},
  {"type": "timeline_estimate", "data_ref": "cert_timeline"}
]
},
{
  "page_number": 11,
  "page_type": "resources",
  "content": {
    "heading": "Learning Resources & References",
    "components": [
      {"type": "course_recommendations", "data_ref": "courses"},
      {"type": "certification_programs", "data_ref": "certifications"},
      {"type": "book_list", "data_ref": "recommended_reading"}
    ]
  }
},
{
  "page_number": 12,
  "page_type": "next_steps",
  "content": {
    "heading": "Next Steps & Contact Information",
    "components": [
      {"type": "action_checklist", "data_ref": "immediate_actions"},
      {"type": "contact_form", "data_ref": "user_contact"},
      {"type": "qr_code", "data_ref": "digital_report_url"},
      {"type": "footer", "data_ref": "branding"}
    ]
  }
}
],

"styling": {
  "fonts": {
    "heading": "Calibri",
    "body": "Arial",
    "mono": "Consolas"
  },
  "colors": {
    "primary": "#2E75B6",
    "secondary": "#00B050",
    "accent": "#FFC000",
    "text": "#333333",
```

```
    "background": "#FFFFFF"
  },
  "page_size": "Letter",
  "orientation": "Portrait",
  "margins": {
    "top": 72,
    "bottom": 72,
    "left": 72,
    "right": 72
  }
}
```

8. REAL-TIME UPDATE PAYLOADS

8.1 WebSocket Progress Update (Payload VIZ-012)

Purpose: Real-time assessment progress updates via WebSocket

```
json
```

```
{
  "payload_id": "VIZ-012",
  "payload_type": "realtime_progress_update",
  "event_type": "assessment_progress",
  "timestamp": "2026-01-03T18:23:47Z",
  "session_id": "session_xyz123",

  "progress_data": {
    "overall_progress": 65,
    "current_phase": "Phase 3: AI Vulnerability Assessment",
    "current_step": "Analyzing protection factors",
    "steps_completed": 42,
    "steps_total": 98,
    "estimated_time_remaining": 180,
    "phase_breakdown": [
      {"phase": "Phase 1: Discovery", "status": "completed", "progress": 100},
      {"phase": "Phase 2: JST Calculation", "status": "completed", "progress": 100},
      {"phase": "Phase 3: Vulnerability", "status": "in_progress", "progress": 45},
      {"phase": "Phase 4: Transferability", "status": "pending", "progress": 0},
      {"phase": "Phase 5: Recommendations", "status": "pending", "progress": 0}
    ]
  },

  "visualization_config": {
    "progress_bar_color": "#2E75B6",
    "completion_color": "#00B050",
    "pending_color": "#CCCCCC",
    "animation_duration": 500,
    "show_percentage": true,
    "show_eta": true,
    "show_phase_details": true
  }
}
```

9. API RESPONSE SCHEMAS

9.1 Complete Assessment Results API Response (Payload VIZ-013)

Purpose: Full API response with all visualization payloads

json

```
{
  "payload_id": "VIZ-013",
  "payload_type": "api_assessment_complete",
  "timestamp": "2026-01-03T18:24:15Z",
  "api_version": "2.0",
  "status": "success",

  "metadata": {
    "user_id": "user_abc123",
    "assessment_id": "assessment_xyz789",
    "processing_time_ms": 12345,
    "token_count": 12936,
    "cost_usd": 0.129
  },

  "results": {
    "jst_index": {
      "payload_ref": "VIZ-001",
      "overall_score": 78,
      "layer_scores": {
        "jobs": 82,
        "skills": 75,
        "talent": 77
      }
    },

    "vulnerability": {
      "payload_ref": "VIZ-004",
      "level": 2,
      "automation_potential": 45,
      "timeline": "3-7 years"
    },

    "transferability": {
      "payload_ref": "VIZ-007",
      "top_opportunities": [
        {"role": "Product Strategy Director", "score": 88},
        {"role": "AI Product Manager", "score": 82},
        {"role": "Chief Product Officer", "score": 75}
      ]
    },

    "recommendations": {
```

```
    "new_skilling": ["Machine Learning", "AI Ethics", "Data Science"],
    "upskilling": ["Executive Communication", "P&L Management"],
    "ready_skilling": ["Claude", "ChatGPT", "Perplexity"]
  }
},

"visualization_payloads": {
  "jst_gauge": "VIZ-001",
  "jst_radar": "VIZ-002",
  "jst_breakdown": "VIZ-003",
  "vulnerability_badge": "VIZ-004",
  "task_heatmap": "VIZ-005",
  "vulnerability_timeline": "VIZ-006",
  "transferability_network": "VIZ-007",
  "skill_gap_matrix": "VIZ-008",
  "dashboard": "VIZ-009",
  "notebooklm_infographic": "VIZ-010",
  "pdf_report": "VIZ-011"
},

"download_links": {
  "pdf_report": "https://api.junglenomics.com/reports/assessment_xyz789.pdf",
  "infographic_png": "https://api.junglenomics.com/infographics/assessment_xyz789.png",
  "data_json": "https://api.junglenomics.com/data/assessment_xyz789.json"
}
}
```

APPENDIX: COLOR PALETTE REFERENCE

json

```

{
  "junglenomics_brand": {
    "primary": "#2E75B6",
    "secondary": "#00B050",
    "accent": "#FFC000",
    "text_primary": "#333333",
    "text_secondary": "#666666",
    "background": "#FFFFFF",
    "background_alt": "#F5F5F5",
    "border": "#CCCCCC"
  },

  "vulnerability_scale": {
    "level_0": {"color": "#FF4444", "bg": "#FFE6E6", "name": "Critical"},
    "level_1": {"color": "#FF8800", "bg": "#FFF0E6", "name": "High"},
    "level_2": {"color": "#FFDD00", "bg": "#FFFACD", "name": "Medium"},
    "level_3": {"color": "#4488FF", "bg": "#E6F0FF", "name": "Low"},
    "level_4": {"color": "#44AA44", "bg": "#E6FFE6", "name": "Resilient"}
  },

  "jst_layers": {
    "jobs": "#2E75B6",
    "skills": "#FFC000",
    "talent": "#00B050"
  },

  "performance_scale": {
    "exceptional": "#44AA44",
    "excellent": "#4488FF",
    "good": "#FFDD00",
    "fair": "#FF8800",
    "needs_development": "#FF4444"
  }
}

```

DOCUMENT CERTIFICATION

Document Type: Visualization Payloads - Complete JSON Data Structures

Platform: ARK SI - Career Intelligence & AI Vulnerability Platform

Total Payloads: 13 primary visualization data structures

Data Format: JSON (production-ready)

Version: 1.0 - January 3, 2026

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Certification: FORGE Ultra Premium (JCSE 48/50)

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END OF VISUALIZATION PAYLOADS DOCUMENT